

Technical Architecture Document

Enterprise Task & Invoice Management System

1. Architecture Overview

- Client-Server Architecture
- RESTful API design
- Role-Based Access Control (RBAC)
- Modular backend structure
- Scalable and secure system design

2. High-Level Architecture Layers

- Presentation Layer (Frontend - React.js)
- Application Layer (Node.js + Express.js)
- Business Logic Layer (Controllers & Services)
- Data Access Layer (Mongoose ODM)
- Database Layer (MongoDB)

3. Backend Architecture

- Node.js runtime environment
- Express.js framework
- MVC pattern implementation
- Middleware for authentication and authorization
- Centralized error handling
- Input validation (Joi/Zod)

4. Authentication & Security

- JWT-based authentication
- Role-Based Access Control middleware
- Password hashing using bcrypt
- Token expiration & refresh strategy
- Rate limiting and request validation
- Soft delete policy for data safety

5. Database Design

- User collection (roles, status, authentication data)
- Workspace collection (assigned users)
- Task collection (workspace reference, assigned user, status, priority)
- Invoice collection (workflow status, items, totals)
- AuditLog collection (user actions tracking)
- Notification collection (optional)

6. Invoice Workflow Engine

- Draft → Pending Approval → Approved/Rejected → Paid
- Approval controlled by Operator/Admin
- Automatic invoice number generation
- PDF generation support
- Audit trail for every status change

7. Logging & Monitoring

- Audit logs for all critical actions
- Login tracking
- System error logging
- Exportable audit reports

8. API Design

- RESTful endpoints
- Versioned API structure (/api/v1)
- Proper HTTP status codes
- Pagination and filtering support
- Swagger API documentation (recommended)

9. Deployment Architecture

- Frontend deployed on cloud hosting (e.g., Vercel/Netlify)
- Backend deployed on cloud server (e.g., Render/AWS)
- MongoDB Atlas cloud database
- Environment-based configuration (.env)
- CI/CD pipeline (optional enhancement)

10. Scalability & Future Enhancements

- Microservices-ready structure
 - Real-time updates (Socket.io)
 - Two-Factor Authentication (2FA)
 - Multi-tenant support
 - Mobile application integration
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Approval

Technical Lead: _____

Signature: _____

Date: _____